

Bachelor AI - MLOps

17 DATA DRIFT

Tobias Mérinat

`tobias.merinat@hslu.ch`

HSLU Hochschule
Luzern

DEPARTMENT OF INFORMATION TECHNOLOGY
Lucerne University of Applied Sciences and Arts
6343 Rotkreuz, Switzerland

September 17/24 and October 15 2026

Software without ML fails due to

- Errors in program logic (bug, wrong requirements)
- Errors in up- or downstream systems (bugs, interface changes)
- Infrastructure problems (downtime, misconfiguration, network problem).

Software fails

- with an exception
- silently

Software with ML can fail in an additional way:

- System runs correctly (none of the above causes)
- but performance still decreases

Cause of Performance Degradation

- Basic assumption in Machine Learning: *Unseen Data* comes from the same distribution as the training data
- During model development, we guarantee this with
 - a fixed training set
 - correct Cross Validation
- Model = Snapshot of reality at the time of training
- But the world is not static, it changes, which means the input data for a model can also change

- **Model Drift:** The circumstance that the prediction quality of an ML model can worsen over time, even though nothing changes in the model itself.
- Synonym: *shift* or simply *drift*
- *Model Drift* is an ill-chosen term, as the model doesn't change, but rather its input data
- **Skew:** Difference between two distributions, especially *Training-Serving-Skew*.
- *Skew* is the difference, *Drift* the (increasing) deviation
- Fundamentally: When the distribution of data at the time of inference differs from that at the time of training

Drift can pose a major problem when we deploy ML in the real world, where data is often dynamic and constantly changing.

- X are the inputs of a model, Y are its outputs
- In Supervised Learning, the training data comes from the joint probability distribution $P(X, Y)$.
- What is modeled is $P(Y|X)$
- For the joint distribution we have
 - $P(X, Y) = P(Y|X)P(X)$
 - $P(X, Y) = P(X|Y)P(Y)$
- Where
 - $P(X)$ is the probability of the inputs
 - $P(Y)$ is the probability of the outputs
 - $P(Y|X)$ is the conditional probability of an output, given the input
 - $P(X|Y)$ is the conditional probability of an input, given the output

Three Types of Drift (Shift)

Drift is divided into three main categories, mixed forms can occur:

- **Covariate Shift** $P(X)$ changes, but $P(Y|X)$ stays the same
- **Label Shift** $P(Y)$ changes, but $P(X|Y)$ stays the same
- **Concept drift** $P(Y|X)$ changes, but $P(X)$ stays the same

Covariate Shift: $P(X)$ changes, $P(Y|X)$ stays the same

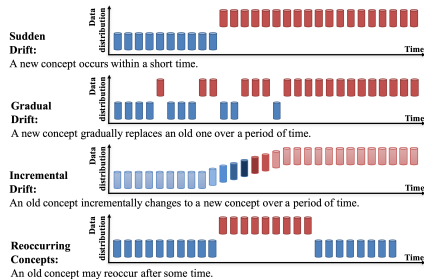
- The inference data is different from the training data
- Can be introduced during training phase, e.g. through Selection Bias
 - Oversampling of an imbalanced dataset
 - Active Learning
- Often triggered by a substantial change in the environment or the way a product is used.
 - App with feature Age is used by a different age group than was represented in the training data
 - or in a different country
 - or through different sensors, newer and higher-resolution cameras, or with noise in the background, while training data was recorded in an isolated room.

Label Shift: $P(Y)$ changes, $P(X|Y)$ stays the same

- Also referred to as *prior shift* or *prior probability shift* or *target shift*
- Distribution of the target variable changes, but for a given label the input distribution stays the same
- Covariate Shift and Label Shift often, but not always, occur together

Concept Drift: $P(Y|X)$ changes, $P(X)$ stays the same

- Also referred to as *posterior shift*
- *Same input, different output*
- Seasonal effects often play a role. Higher prices on weekends than during the week.
- Can come gradually or suddenly

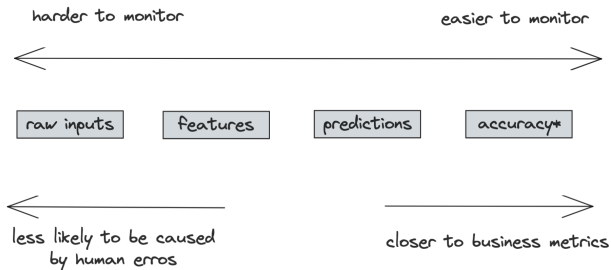


- Input data and model performance must be monitored
- Models must be retrained periodically

Monitoring Drift

We can distinguish four levels of drift monitoring

- 1 Raw Data
- 2 Features
- 3 Predictions
- 4 Metrics



* if natural labels available

Metrics (Natural Labels)

- To measure metrics directly, we need *Natural Labels*
- We store the predictions from inference and calculate and check periodically
- The faster the labels arrive, the faster we can react
- Besides Natural Labels, any other feedback that could serve as a proxy for a Natural Label should also be logged.
 - Clicks on Recommendations
 - Bookmarks
 - Upvotes
 - Shares
 - Likes
 - Click on *skip* or *abort* (Spotify, Youtube)
 - Usage duration
 - . . .
- Such values can be used to hint at occurring changes

- If we don't have fresh Ground Truth (Natural Labels), we compare distributions of current data with distributions of reference data
 - 1 We form windows over the inference requests
 - 2 calculate the distribution of a window
 - 3 use a metric to compare the calculated distribution with a reference dataset
 - 4 store the calculated metrics as time series to identify trends
- Features: simpler, because schema; possibility to monitor intermediate stages besides final values
- Raw data: same procedure, but somewhat more difficult due to often missing structure
- Predictions: If a Natural Label is missing, one can also measure changes in the distribution of predictions

How can text data drift manifest itself? A few Examples:

- Words change their meaning: The word *sick* slowly changes from a negative to a positive sentiment
- New popular words, brands, abbreviations or memes emerge: *tl;dr*, *LOL*, *BRB*, *GDPR*, *tiktok*, *deepseek*, . . .
- New ways of writing, e.g. mixing languages, *Spanglish*: *I'm so tired lah, trabajo all day sia*
- Twitter removes 140 character limit, texts become longer

Drift Metrics for Text Data 1/2

■ Text statistics

- Average / min / max # of chars / words / sentences / paragraphs

■ Text health

- OOV word percentage (based on given vocab)
- Non-Letter character percentage

■ Text content (avg number of matches / percentage of texts with at least one)

- special markers (hashtags, @mentions, emojis, links)
- special matches (begins with, contains, regex, . . .)
- trigger words

	current	reference
count	4929	4805
missing	71 (0.014%)	195 (0.039%)
Text Length min	0.0	0.0
Text Length mean	311.571	290.544
Text Length max	508.0	504.0
OOV % min	0.0	0.0
OOV % mean	5.853	5.378
OOV % max	40.0	42.857
Non Letter Character % min	0.0	0.0
Non Letter Character % mean	3.629	3.802
Non Letter Character % max	12.17	16.667

Source: Evidently

- Classifier-based
 - average sentiment score
 - counts by classes
 - similarity scores to reference
- Embeddings Drift (with or without prior dimensionality reduction such as PCA or T-SNE)
 - Distance metrics (euclidean, cosine, . . .)
 - Classifier model
 - Share of drifted embedding components (treat embedding as a table)

Which metric to choose?

- How representative is it for the use case?
- How easy is it to interpret the drift score?
- How stable are the results? How much variance is there?
- How fast are the computations?
- How fragile is the method?

If we think that the drift impairs model performance too much, the model must be retrained.

- Train only with new data (label if necessary)
- Train with old and new data
- Test which old data is still good and can be kept
- Evaluate and compare with old model

- In general, monitor as much as possible and continuously adjust
- Brainstorm what could go wrong, then define appropriate metrics
 - Operational Metrics: Memory, CPU utilization, latency, throughput, server load
 - Statistical Health of input data
 - Number of NaNs, Average Image Brightness
 - Statistical Health of outputs
 - Distribution, number of nulls, number of True in a row
 - User behavior: CTR, time spent with content, likes, shares, upvotes, . . .
 - Performance on a particularly important data subset (as a separate metric)

- Data types
- Data ranges (min, max or value set)
- Regexps
- Not Null
- Key Constraints